

**NATIONAL UNIVERSITY OF SCIENCES &
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Project Report: DLD

**Design and Implementation of a Line Following
Robot Using Logic Gates**

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1 Abstract

This project presents the design and implementation of a line-following robot using digital logic components. The robot autonomously follows a predefined path using three sensors and a combinational logic circuit. Unlike conventional systems that rely on microcontrollers, this design strictly uses logic ICs (7404, 7432, 7408) and an L293D motor driver. The objective is to demonstrate how real-world automation problems can be solved using fundamental Digital Logic Design concepts. The system was simulated in Proteus and later implemented in hardware using a PCB. The robot successfully detects path variations and adjusts its motion accordingly.

2 Introduction

Digital Logic Design is the backbone of modern electronic systems. In industrial environments, automation is essential for improving efficiency and reducing manual labor. One common automation problem is transporting materials from one location to another. A practical solution is a line-following robot, which follows a predefined path marked on the ground. This project explores the implementation of such a robot using basic logic gates instead of programmable controllers, encouraging a deeper understanding of hardware-level decision-making.

3 Objectives

The main objectives of this project are:

- To design a fully functional line-following robot
- To implement decision-making using combinational logic circuits
- To avoid the use of programmable devices (as per PBL constraints)
- To analyze and simulate the circuit before hardware implementation
- To understand real-world applications of DLD concepts

4 Literature Background

A line-following robot typically uses sensors to detect contrast between a line and the surrounding surface. Most modern designs use microcontrollers like Arduino for processing. However, this project follows a pure hardware logic approach, using:

- Boolean algebra
- Logic gates
- Combinational circuit design

This approach is important because it builds strong fundamentals, removes dependency on software, and demonstrates hardware efficiency.

5 System Overview

The robot system consists of four major subsystems:

5.1 Input System (Sensors)

Two IR sensors detect the path:

- Left (L)
- Right (R)

One is for counting junctions:

- Junction sensor (J)

5.2 Processing Unit (Logic Circuit)

Implemented using:

- NOT gates (7404)
- OR gates (7432)
- AND gates (7408)
- Decade Counter (4017)

This unit decides robot movement based on sensor input.

5.3 Control Unit (Motor Driver)

- L293D IC: Converts logic signals into motor control signals.

5.4 Output System (Motors)

- Two DC motors: Enable robot movement and turning.

6 Detailed Working Principle

6.1 Sensor Behavior

Each sensor produces digital output:

- White surface $\rightarrow$ Logic 1
- Black line $\rightarrow$ Logic 0

6.2 Decision Logic

The robot must react based on line position:

- Line in center $\rightarrow$ move forward
- Line left $\rightarrow$ turn left
- Line right $\rightarrow$ turn right

6.3 Logic Implementation

- **Step 1: Signal Inversion (7404):** Some conditions require inverted signals: $\text{NOT}(L)$, $\text{NOT}(J)$, $\text{NOT}(R)$.
- **Step 2: Condition Formation (7432 OR Gates):** OR gates combine conditions such as detecting multiple sensor combinations.
- **Step 3: Final Decision (7408 AND Gates):** AND gates ensure correct combination of signals for motor control inputs.

6.4 Boolean Logic Representation

The movement Boolean functions are represented as follows:

- **Forward:** $F = \bar{L} \cdot J \cdot \bar{R}$
Robot moves forward when only the center/junction sensor detects the line.
- **Turn Left:** $TL = L \cdot \bar{J} \cdot \bar{R}$
Robot turns left when the left sensor detects the line.
- **Turn Right:** $TR = \bar{L} \cdot \bar{J} \cdot R$
Robot turns right when the right sensor detects the line.
- **Stop/Junction Detected:** $S = L \cdot J \cdot R$
Robot stops when all sensors detect black, meaning it has reached a junction.

7 Truth Table

The simplified sensor-to-movement logic table is detailed below:

L	R	Movement
0	0	Forward
1	0	Right
0	1	Left
1	1	Forward

8 Motor Driver (L293D) Working

The L293D is a dual H-bridge motor driver featuring:

- Direction control of motors
- Bidirectional rotation capabilities
- Current amplification

8.1 Pin Functionality

- IN1, IN2 $\rightarrow$ Motor A
- IN3, IN4 $\rightarrow$ Motor B
- EN pins $\rightarrow$ Enable motors

8.2 Motor Logic Example

IN1	IN2	Motor State
1	0	Forward
0	1	Reverse
0	0	Stop

9 Circuit Design Explanation

From simulation, the system workflow operates through a layered hardware hierarchy:

1. Sensors are connected to NOT gates.
2. Outputs go into OR gates.
3. Then processed by AND gates.
4. Final outputs go to L293D inputs.
5. Motors respond accordingly.

This layered approach ensures a clean logic design, proper signal conditioning, and accurate movement.

10 Simulation

The circuit was simulated using Proteus software.

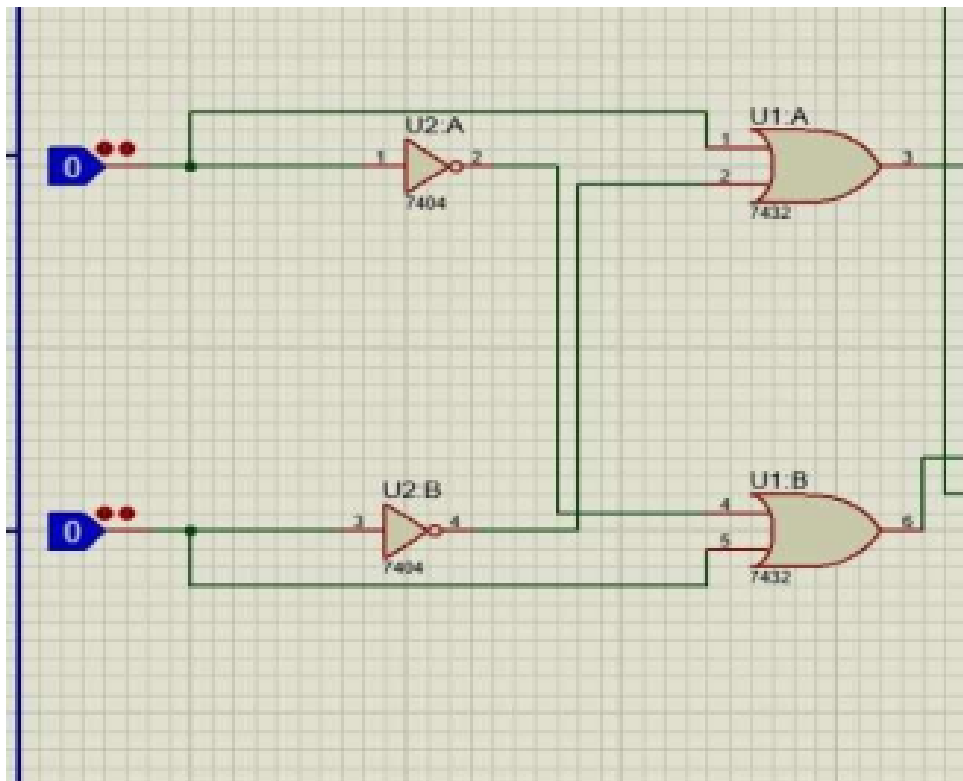


Figure 1: Line Following Logic

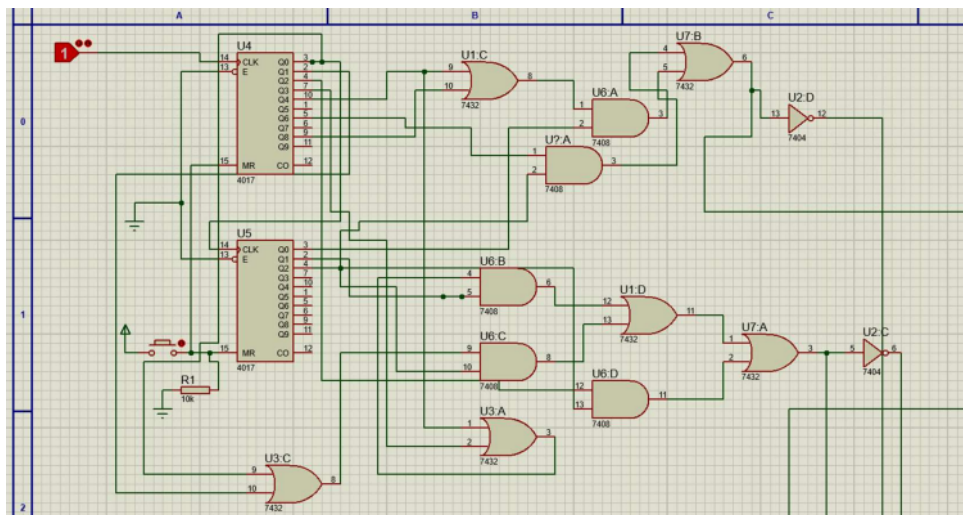


Figure 2: Junction Logic

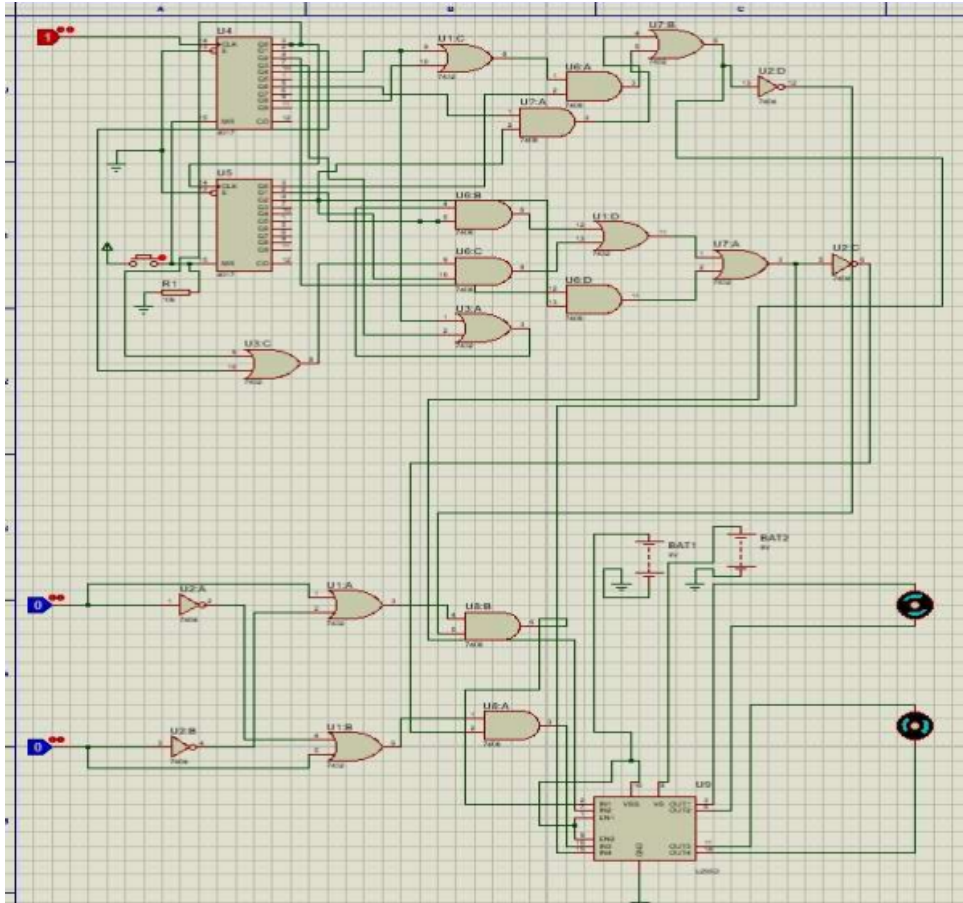


Figure 3: Proteus Simulation Full Logic

10.1 Simulation Results

- Verified logic conditions
- Tested all sensor combinations
- Ensured correct motor behavior
- Debugged wiring errors

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11 Hardware Implementation

11.1 Assembly

- All ICs mounted securely on a breadboard.
- Sensors aligned properly.
- Motors connected via L293D.
- Battery integrated.

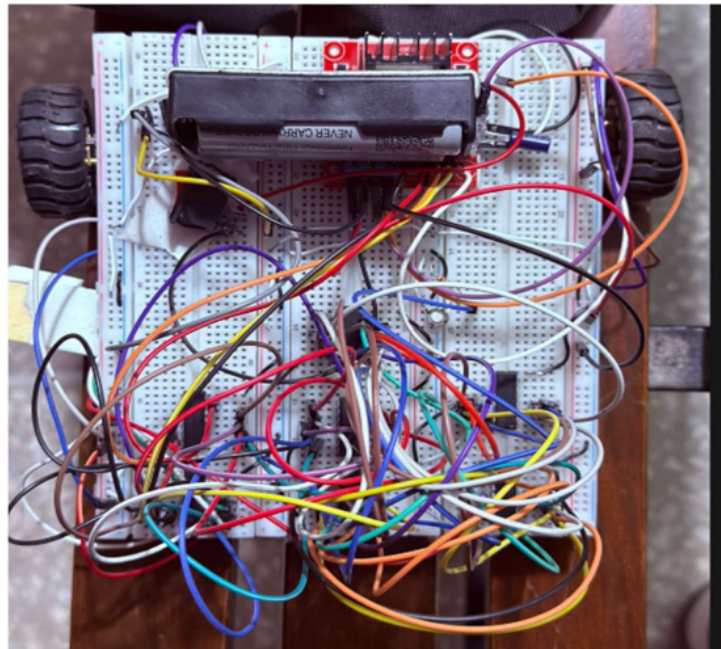


Figure 4: Hardware Implementation

11.2 Testing

- Robot tested on track
- Verified turning accuracy
- Adjusted sensor alignment

12 Challenges Faced

- Logic simplification without a microcontroller
- Noise in sensor readings
- Wiring complexity
- Motor synchronization

13 Advantages

- Fully hardware-based design
- No programming required
- Fast response time
- Educational value

14 Limitations

- Less flexibility
- Hard to modify logic
- Limited scalability

15 Applications

- Industrial automation
- Warehouse robots
- Delivery systems
- Educational robotics

16 Future Enhancements

- Add PWM speed control
- Use additional sensors
- Integrate obstacle detection
- Combine with AI/microcontroller

17 Work Distribution

Member	Responsibility
Nouman	Logic Design & Boolean Equations
Sarim	Proteus Simulation
Tahir	Breadboard Implementation
Saad Akbar	Hardware Assembly
Huzaifa	Testing & Debugging
Hamza Babar	Documentation & Report

18 Conclusion

This project successfully demonstrates how a line-following robot can be built using digital logic circuits only. It highlights the practical application of combinational logic and reinforces the importance of hardware-level design.

The robot performs efficiently and meets all requirements defined in the PBL guidelines. This project enhances both theoretical understanding and practical engineering skills.